Частина 2:

Варіант 17.

+ 1)

а) $\left(\frac{1+i}{1-4i}\right)^2 = \frac{1+2i+i^2}{1-8i+16i^2} = \frac{1-1+2i}{1-8i-16} = \frac{2i}{-8i-15} = \frac{-2i}{8i+15} =$

$= \frac{-2i(8i-15)}{-64-225} = \frac{+2i(8i-15)}{289} = -\frac{16}{289} - \frac{30i}{289}$

б) $\sqrt[4]{\sqrt{3}+i}$

$|z| = \sqrt{\sqrt{3}^2+1^2} = \sqrt{3+1} = 2$

$\arg(z) = \operatorname{arctg}\frac{1}{\sqrt{3}} = \frac{\pi}{3}$

$\sqrt[4]{\sqrt{3}+i} = 2\left(\cos\left(\frac{\pi}{12}\right) + i\sin\left(\frac{\pi}{12}\right)\right)$

$= 2\left(\cos\left(\frac{7\pi}{12}\right) + i\sin\left(\frac{7\pi}{12}\right)\right)$

$= 2\left(\cos\left(\frac{13\pi}{12}\right) + i\sin\left(\frac{13\pi}{12}\right)\right)$

$= 2\left(\cos\left(\frac{19\pi}{12}\right) + i\sin\left(\frac{19\pi}{12}\right)\right)$

+ 2) $\lim\limits_{n\to\infty} \frac{\sqrt{n^8+6} - \sqrt{n-6}}{\sqrt[5]{n^8+6} + \sqrt{n-6}} = \lim\limits_{n\to\infty} \frac{n^4\left(\sqrt{1+\frac{6}{n^8}}\right) - \sqrt{n}\left(\sqrt{1-\frac{6}{n}}\right)}{n\sqrt[5]{n^3}\left(\sqrt[5]{1+\frac{6}{n^8}}\right) + \sqrt{n}\left(\sqrt{1-\frac{6}{n}}\right)} =$

$$= \lim_{n\to\infty} \frac{n^3\sqrt{n}\left(\sqrt{1+\frac{6}{n^8}}\right) - 1\left(\sqrt{1-\frac{6}{n}}\right)}{\sqrt[4]{n}\sqrt[5]{n^3}\left(\sqrt{1+\frac{6}{n^8}}\right) + 1\left(\sqrt{1-\frac{6}{n}}\right)} =$$

$$= \lim_{n\to\infty} \frac{\sqrt[4]{n^{12}}\left(\underset{=1}{\sqrt{1+\frac{6}{n^8}}}\right) - \overset{\to 0}{\frac{1}{\sqrt[4]{n^3}}}\left(\sqrt{1-\frac{6}{n}}\right)}{\underset{=1}{\sqrt{1+\frac{6}{n^8}}} + \underset{\to 0}{\frac{1}{\sqrt[4]{n^3}}}\left(\sqrt{1-\frac{6}{n}}\right)} = \infty$$

+ 3) $$\lim_{n\to\infty} \frac{1+4+7+\dots+3n-2}{\sqrt{5}\,n^4+n+1} = \lim_{n\to\infty} \frac{3n^2+n}{2\sqrt{5}n^4+2n+2} =$$

$$= \lim_{n\to\infty} \frac{n^2\left(3+\frac{1}{n}\right)}{n^4\left(2\sqrt{5}+\frac{2}{n^3}+\frac{2}{n^4}\right)} = \lim_{n\to\infty} \frac{\overset{\to 3}{3}+\frac{1}{n}}{\underset{\infty}{n^2}\left(\underset{2\sqrt{5}}{2\sqrt{5}}+\underset{\to 0}{\frac{2}{n^3}}+\underset{\to 0}{\frac{2}{n^4}}\right)} = 0$$

+ 4) $$\lim_{n\to\infty} \left(\frac{2n^2+18n-7}{2n^2+17n+9}\right)^{n+1} = \lim_{n\to\infty} \left(1+\frac{n-16}{2n^2+17n+9}\right)^{n+1} =$$

$$= \lim_{n\to\infty} \left(\left(1+\frac{1}{\frac{2n^2+17n+19}{n-16}}\right)^{\left(\frac{2n^2+17n+19}{n-16}\right)\left(\frac{n-16}{2n^2+17n+19}\right)}\right)^{(n+1)} =$$

$$= \lim_{n\to\infty} e^{\lim\limits_{n\to\infty}(n+1)\left(\frac{(n-16)}{2n^2+17n+19}\right)} = e^{\lim\limits_{n\to\infty}\frac{n^2\left(1-\frac{15}{n}-\frac{16}{n^2}\right)}{n^2\left(2+\frac{17}{n}+\frac{19}{n^2}\right)}} = e^{0,5}$$

+ 5) $$\lim_{x\to-1} \frac{x^3-3x-2}{(x^2-x-2)^2} = \lim_{x\to-1} \frac{(x+1)^2(x-2)}{(x+1)^2(x-2)^2} = \lim_{x\to-1} \frac{1}{x-2} = -\frac{1}{3}$$

+ 6) $$\lim_{x\to 0} \frac{\sqrt{x+1}-\sqrt{1-x}}{\sqrt[3]{x}} = \lim_{x\to 0} \frac{2x}{\sqrt[3]{x}\left(\sqrt{x+1}+\sqrt{1-x}\right)} =$$

$$\lim_{x\to 0} \frac{2\sqrt{x}}{\sqrt[4]{x}\left(\left(\sqrt{1+\frac{1}{\sqrt[4]{x}}}\right)+\left(\sqrt{1-\frac{1}{\sqrt[4]{x}}}\right)\right)} = \lim_{x\to 0} \frac{2\sqrt[4]{x^5}}{\sqrt{1+\frac{1}{\sqrt[4]{x}}}+\sqrt{1-\frac{1}{x}}} = 0$$

7) $y = \frac{\cos 2x}{1-\cos x}$

1. ОДЗ:

$\cos x \neq 1$

$x \neq 0 + 2\pi k,\ k \in Z$

2. $f(x) = \frac{\cos 2x}{1-\cos x}$

$f(-x) = \frac{\cos(-2x)}{1-\cos(-x)} = \frac{\cos 2x}{1-\cos x}$ $\Rightarrow$ Парна

3. $T = 2\pi$

4. $y = 0 \Rightarrow \cos 2x = 0 \quad x = \pm\frac{\pi}{4} + 2\pi k,\ k \in Z$

$x = 0 \Rightarrow y = \frac{0}{0};\ x \notin$ ОДЗ

5. $\lim_{x\to 0-0} \frac{\cos 2x}{1-\cos x} = \lim_{x\to 0-0} \frac{\cos(-2\varepsilon)}{1-\cos(-\varepsilon)} = \lim_{x\to 0-0} \frac{1-2\sin^2(-\varepsilon)}{-2\sin 4(-\frac{\varepsilon}{2})} = \lim_{x\to 0-0} \frac{1-2\varepsilon^2}{-\varepsilon^2} = \infty$

$\Downarrow$

2го род

6) $y = \frac{\cos(2x)}{1-\cos x}$

$-2\sin 2x - \cos x \sin x$

$$y' = \frac{-2\sin(2x)(1-\cos x) - (\cos(2x))\cdot(\sin x)}{(1-\cos x)^2} = \frac{2\cos x \sin 2x \text{ [illegible]}}{(1-\cos x)^2}$$

$$y' = \cancel{\sin x}\ \frac{4\cos^2 x \sin x - 4\cos x \sin x - \cos 2x \sin x}{(1-\cos x)^2} =$$

$$= \sin x \left(\frac{4\cos^2 x - 4\cos x - \cos 2x}{(1-\cos x)^2} \right) =$$

$$= \frac{\sin x \left(4\cos^2 x - 4\cos x - 2\cos^2 x + 1 \right)}{(1-\cos x)^2} =$$

$$= \frac{\sin x}{(1-\cos x)^2} \left(2\cos^2 x - 4\cos x + 1 \right)$$

$2t^2 - 4t + 1 = 0$

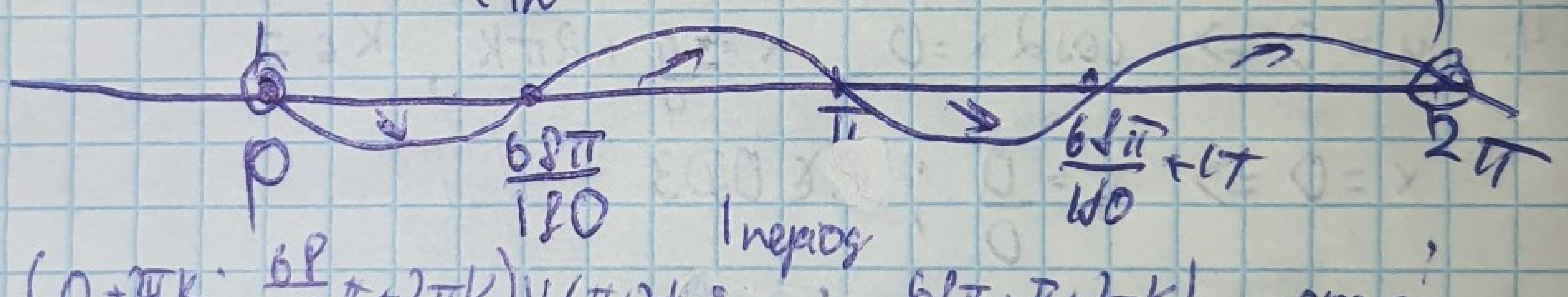
$D = 16 - 8 = 8$

$t = \frac{2 \pm \sqrt{2}}{2} \Rightarrow$ $t_1 = 1 + \frac{\sqrt{2}}{2} > 1$ $\boxed{-}$

$t_2 = 1 - \frac{\sqrt{2}}{2} \Rightarrow x \approx 68^\circ \approx \frac{68}{180}\pi$

$x = \frac{\pi}{2}$, $x = 0$

$x \in \left(\frac{68}{180}\pi + 2\pi k;\ \pi + 2\pi k\right) \cup \left(\frac{68\pi}{180} + \pi + 2\pi k;\ 2\pi k + 2\pi k\right)$ $k \in Z$



$x \in \left(0 + \pi k;\ \frac{68}{180}\pi + 2\pi k\right) \cup \left(\pi + 2\pi k;\ \frac{68\pi}{180} + \pi + 2\pi k\right)$ — спадає

$k \in Z$

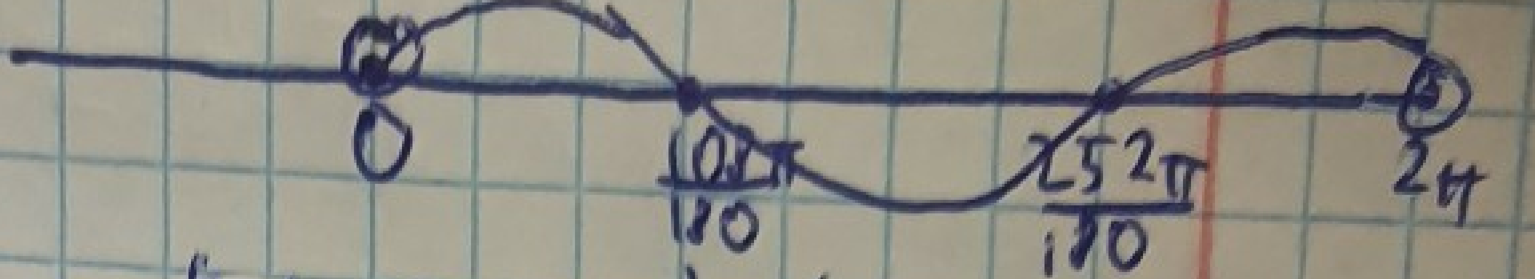
7. $y'' = \left(\frac{\sin x}{(1-\cos x)^2} (2\cos^2 x - 4\cos x + 1) \right)' =$

$$= \frac{\cos x (1-\cos x)^2 - 2(1-\cos x)(+\sin x \cdot \cos x)}{(1-\cos x)^4} (2\cos^2 x - 4\cos x + 1)$$

$$+ (-4\cos x \sin x + 4\sin x) \left(\frac{\sin x}{(1-\cos x)^2} \right) = 0$$

$x = \frac{108\pi}{180} + 2\pi k$, $k \in Z$

$x = \frac{252\pi}{180} + 2\pi k$, $k \in Z$



$x \in \left(2\pi k; \frac{108\pi}{180} + 2\pi k\right) \cup \left(\frac{252\pi}{180} + 2\pi k; 2\pi + 2\pi k\right)$ – оп вниз

8.) 3 5) : $x = 0 + 2\pi k$ – верт ас.

$x \in \left(\frac{108\pi}{180} + 2\pi k; \frac{252\pi}{180} + 2\pi k\right)$ – оп вгору $k \in Z$

$\lim\limits_{x \to \pm\infty} \frac{\cos 2x}{1 - \cos x}$ – невизначена → гор ас немає.

$\lim\limits_{x \to \pm\infty} \frac{\cos 2x}{x(1 - \cos x)} = 0 \Rightarrow$ [illegible]

$\lim\limits_{x \to \pm\infty} \frac{\cos 2x}{(1 - \cos x)} \Rightarrow$ невизн → похилих ас. нема: